

Animating with Battle Cats Ultimate

Version 1.1

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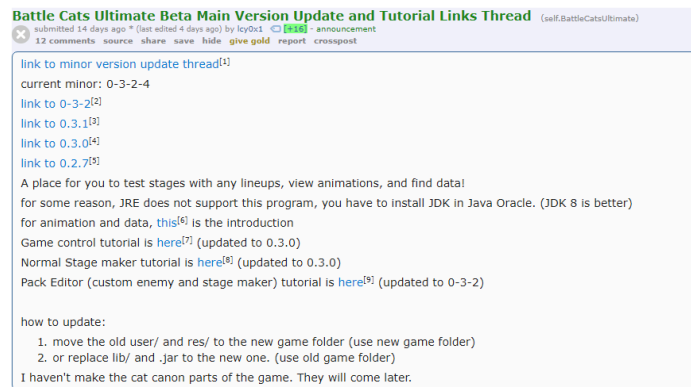
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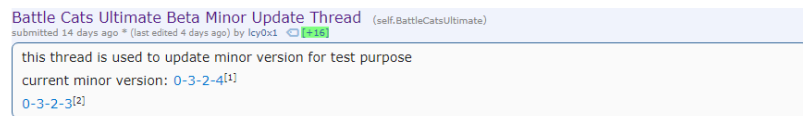
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Part 1 - Downloading the software

To download the software as of 09-07-2018, go to the [Battle Cats Ultimate subreddit](#) (or go to the discord linked on the subreddit) and download the current major version and minor version of the program. You will need to have Java to run this program.



[Major Download Link](#)



[Minor Download Link](#)

Extract the major version with WinRar or 7zip and place the minor .jar in the top level of the extracted folder.

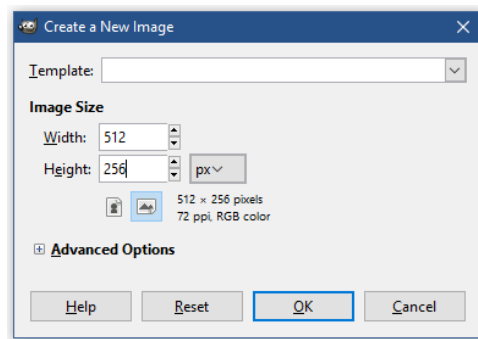
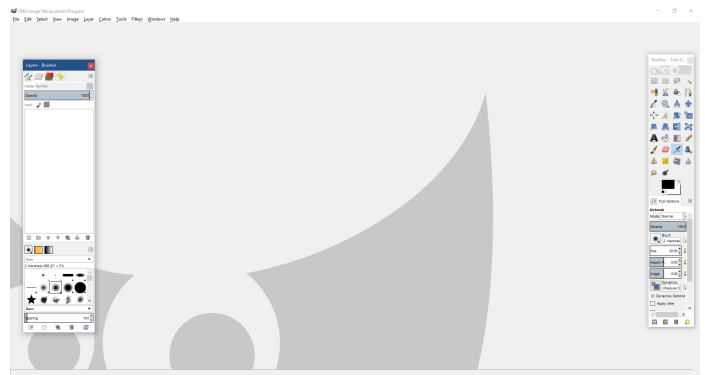
Once you're done with that, go into the BCUB/res/anim and create a folder with the name of your unit. I will use CatMoji for my animation.

**** i have no idea what the process is anymore lol, if you're reading this, i assume you've already installed BCU****

Part 2 - Creating your sprite-sheet

To get started with your sprite-sheet, I recommend you to download [Gimp](#) or [Paint.net](#) as a free image editing software, but for this guide, I will use Gimp I find it more user friendly.

Starting up gimp will open up 3 windows, but for now what you want to do is go to the top left, click File, and then New.



A new window should pop up asking if you want to change your image size. Most Battle Cats sprite-sheets are 512 x 256 px. It can be as big as you want it, but smaller sizes will be easier to measure and cut. Change the size and hit Ok.



Gimp will open up a blank canvas with the borders defined by a dotted yellow line. You can start drawing on this if you want to, but we will need to remove the background first.

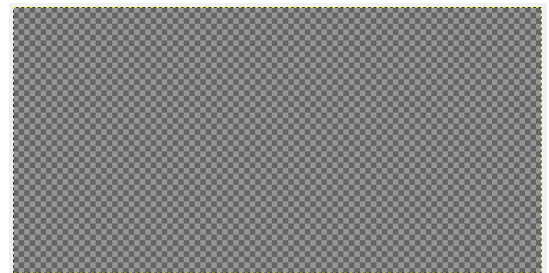


If you aren't already on it, open up the layers tab on the left window (The stack of 3 papers). Delete that layer by hitting the trash can icon in the middle of the window, and create a new layer with the paper icon on the same row. Change layer type to transparency if it's not already on it and click OK.

Layer Fill Type

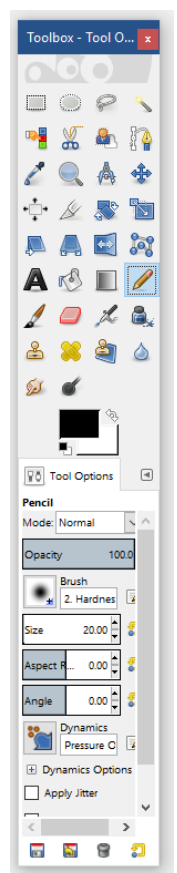
- ☐ Foreground color
- ☐ Background color
- ☐ White
- ☒ Transparency

This will give you a checkerboard pattern on your canvas, but this will not show on your final image. It is only there to show that the image is transparent and that nothing is currently drawn on it.



Over on the right, you have your Tool Box and Tool options. It may appear to be complicated, but for basic drawing and editing, you only need to know a few tools. Hovering over the icons will give you the name of the tool.

Rectangle Select Tool - Can select in rectangular shapes



Free Select Tool - Can select in any shape, click to create the outline for your selection, and join the ends together to complete the selection.

Adding to your selection can be done by holding Shift while selecting

If you can't remember keyboard shortcuts, right-click on the canvas, go to edit, and you can Cut, Copy, Copy Visible, and Paste.

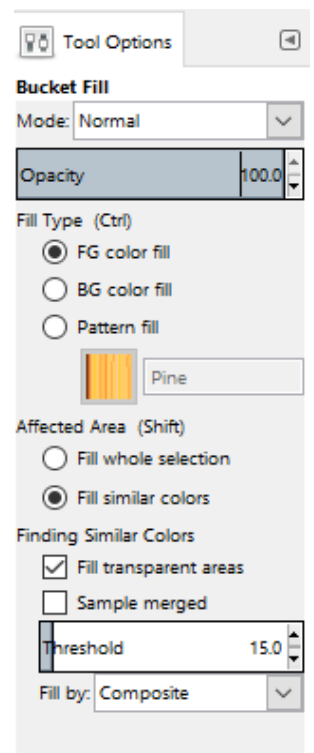
If you paste an image, to confirm placement, you must have a selection tool selected.

Measure Tool - Gives (x,y) values for points and can measure lengths. Displays information at bottom of Canvas Window.

Rotate Tool - Select an image with a selection tool, then select the Rotate Tool. Click on the canvas and a window will open up, allowing you to rotate either by moving the slider, or dragging the corners of your image.

Scale Tool - Select an image with a selection tool, then select the Scale Tool. Click on the canvas and a window will open up, allowing you to expand your image by changing the x and y values or grabbing the corners of your image. Clicking the chain will lock and unlock the function of keeping the proportions of your image.

Bucket Fill Tool - Fill an enclosed area with color. To edit harshness of fill, scroll down in tool options to threshold and raise or lower the value

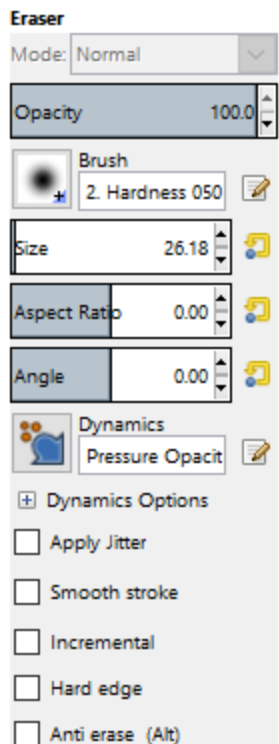


Pencil Tool - Use this to draw with only solid pixels

Paintbrush Tool - Use this to draw with pixels of varying opacity

Eraser Tool - Use this tool to erase pixels. Toggle Hard edge to make your eraser sharper

All three of these tools share the same Tool Options, except for Eraser which has a few more.



Opacity - Changes how transparent your tool is

Size - Changes the size of your tool

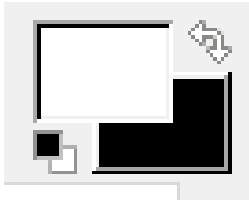
Playing around with dynamics will let you be able to use pen-pressure but I haven't used those yet.

[Read this if you want to find out how to do this.](#)

Scrolling in and out will zoom in and out on the canvas.

Holding down space while moving the mouse will move you around the canvas.





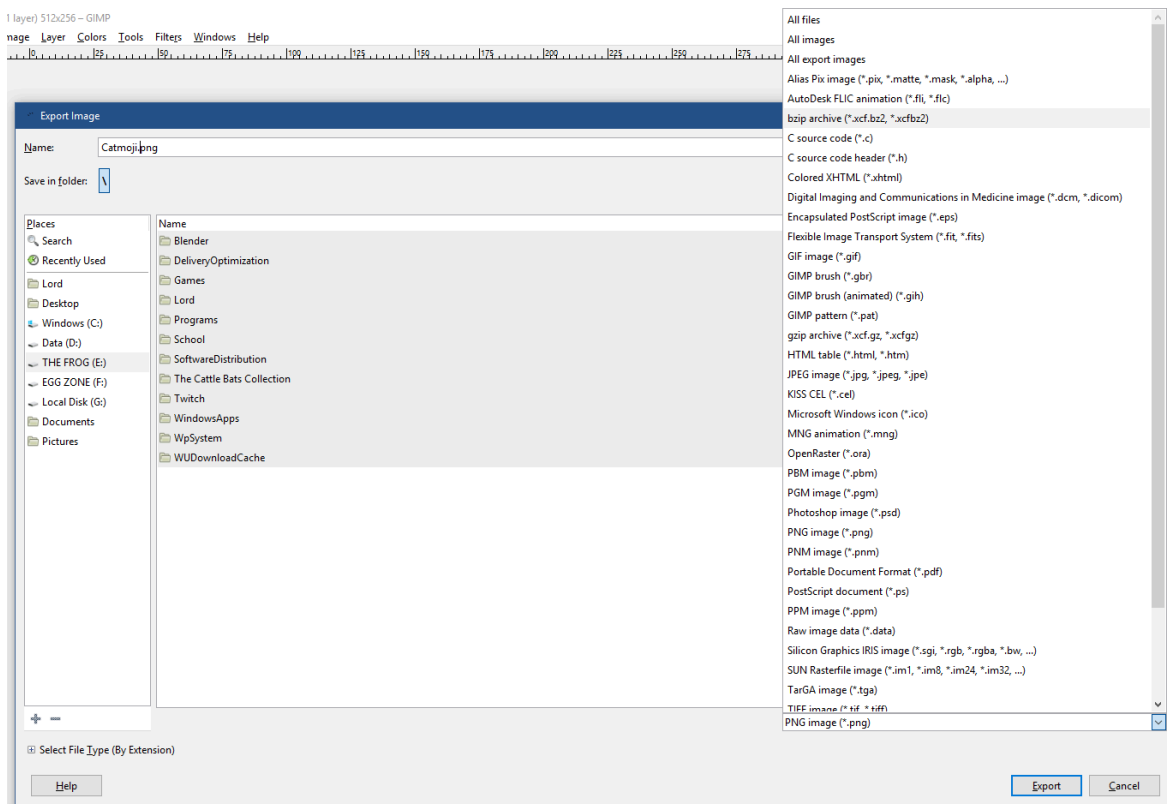
Clicking on a box will switch you to that color.

Clicking on the arrows will swap the 2 colors around.
Useful for quick-swapping

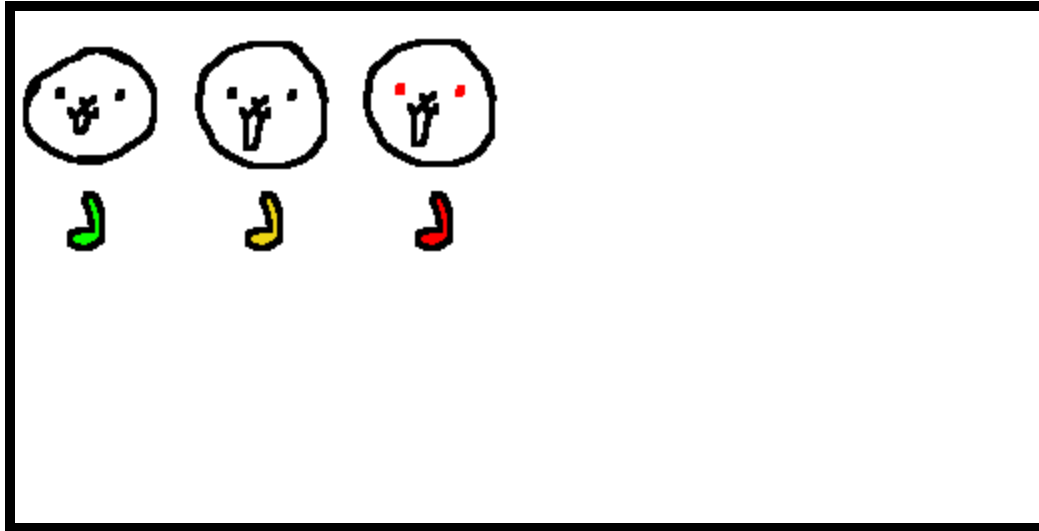
The 2 small boxes will set the colors back to white and black.

Double Clicking on a box will open up the color selector.

To export your image, go to File, then Export.



Change “All files” to PNG image, and export to your named folder.



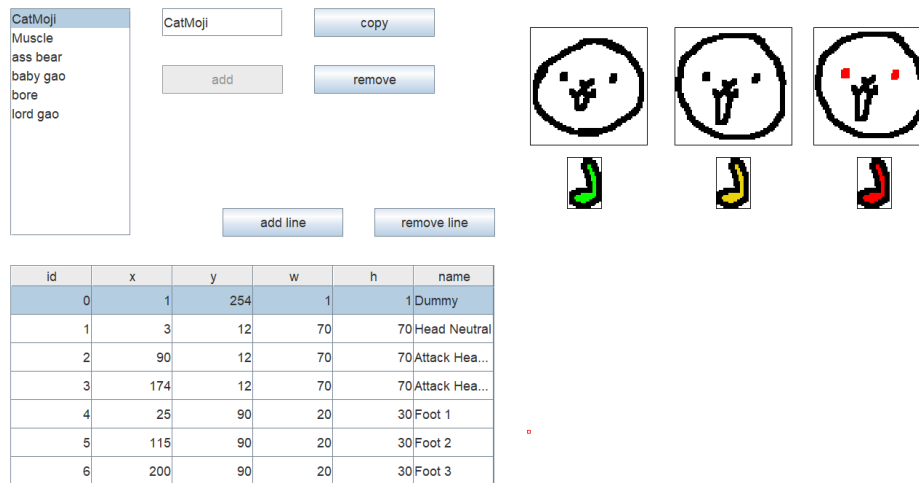
This is my simple sprite-sheet. Depending on whether you want to do frame by frame animation (look at Basic Unit sprite-sheets), or puppet rig animation (look at most Uber sprite-sheets) you might have more or less on your image.



Part 3 - Creating your .imgcut

Now that you have your sprite-sheet created, open up the newest version of BCU and open edit imgcut.

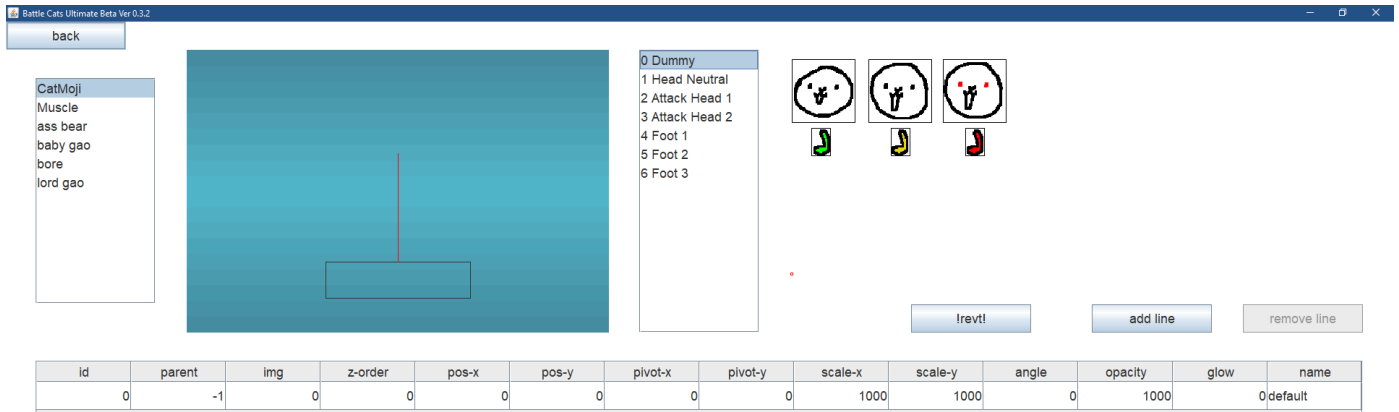
Here is where you need to use the Measure Tool in Gimp. Find the part you want to measure and determine the top left corner of that part. This will be the x and y value of the part. Click and drag from that point to determine the width and height of that part. That will be the w and h values. Then, name the part whatever you want to help you remember. Repeat for all parts. If you want, you can eyeball each piece and keep plugging in values until it works, but making sure to measure correctly will help in the long run. Outlines will appear whenever you fully input a part.



id	x	y	w	h	name
0	1	254	1	1	Dummy
1	3	12	70	70	Head Neutral
2	90	12	70	70	Attack Hea...
3	174	12	70	70	Attack Hea...
4	25	90	20	30	Foot 1
5	115	90	20	30	Foot 2
6	200	90	20	30	Foot 3

Note: Something that I have observed from official animations is that PONOS adds a “Dummy” part with nothing inside as the part with id 0. This helps with another part later.

Part 4 - Creating your .mamodel



Enter this by going back to the menu and going to Edit mamodel.

Let's explain the different values

id - This is what you will refer to later whenever you are animating. It's automatically filled out for you.

Parent - This value determines what a part's parent is. If it's -1 then the part does not have a parent. If it does have a parent, the parent value of the child will be the same value as the id value of the parent part. The child's pivot will be on the same point as the parent's pivot without any transformations.

img - This value determines what part it uses from .imgcut. The values are displayed on the side of the preview.

z-order - Z-order determines when a part is drawn. A part with a higher z-order value will be on top of one with a lower value

pos-x - Moves a part relative to its parent's pivot along the x-axis.

pos-y - Moves a part relative to its parent's pivot along the y-axis.

pivot-x - Moves a part's pivot point on itself along the x-axis If you want a part to rotate on a certain point, you would edit this value

pivot-y - Moves a part's pivot point on itself along the y-axis If you want a part to rotate on a certain point, you would edit this value

scale-x - Makes a part wider or thinner. Neutral value is 1000.

scale-y - Makes a part taller or shorter. Neutral value is 1000.

Angle - Rotates a part clockwise. Max value is 3600. Can also be negative to rotate counter-clockwise.

Opacity - Controls how transparent a part is. Max value is 255

Glow - Glow can either be 0 or 1. 0 means no glow, and 1 means that it will remove all black from a part. Looking at Cosmo's sprite-sheet, you can see that some of the glowing effects are surrounded in black. With glow, all black is removed from the part, making the effect transparent.

Honestly, you don't need this as you can just make your part transparent without the black background.



Name - Nothing important, just make sure to name things properly for convenience. Note that the names of parts from copied animations are usually a string of question marks, so renaming them manually may be helpful.

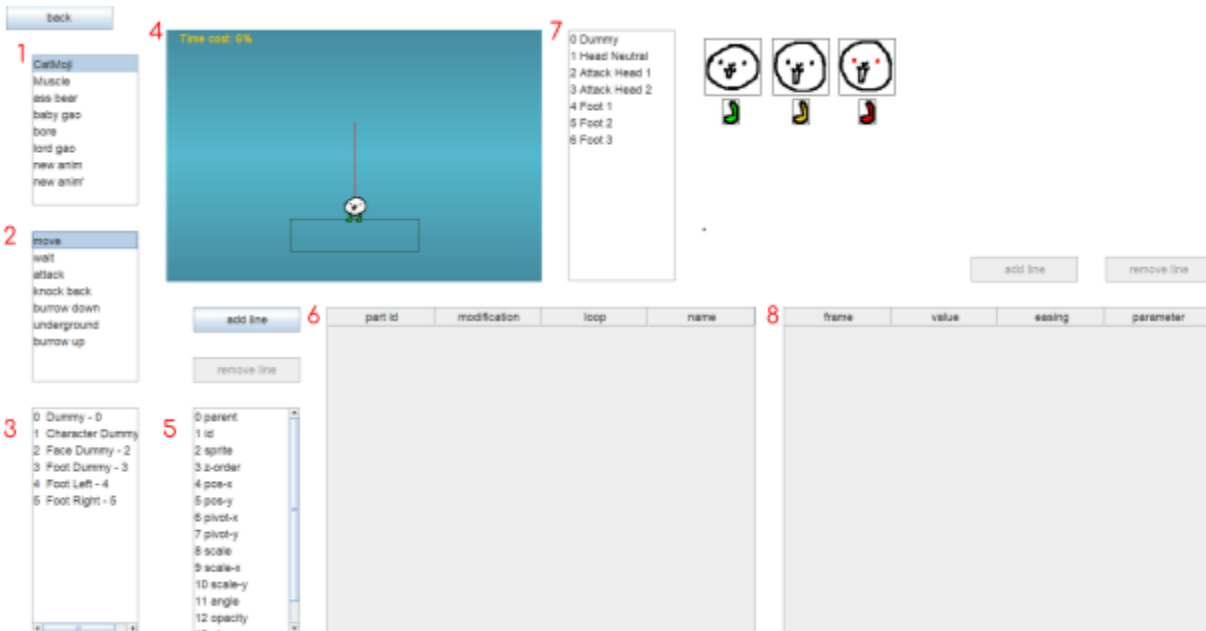
Similar to .imgcut, .mamodel uses the dummy part that we made. This dummy will be the parent to all our parts so we can move the entire unit around. Other parts that you want to keep together should also use the dummy image model.

id	parent	img	z-order	pos-x	pos-y	pivot-x	pivot-y	scale-x	scale-y	angle	opacity	glow	name
0	-1	0	0	0	0	0	0	1000	1000	0	255	0	Dummy - 0
1	0	0	0	3	0	0	0	1000	1000	0	255	0	Character Du...
2	1	1	1	0	-40	35	35	1000	1000	0	255	0	Face Dummy...
3	0	0	0	0	-15	0	0	1000	1000	0	255	0	Foot Dummy ...
4	3	4	2	-15	0	10	5	1000	1000	0	255	0	Foot Left - 4
5	3	4	3	15	0	10	5	1000	1000	0	255	0	Foot Right - 5



Part 5 - Creating your .maanim

Enter this by going back to the menu and going to Edit maanim.



At first, the viewer will be blank, but after selecting your animation in (1), 3 things will appear. The unit's sprite-sheet will appear in (7) along with the names and ids that you gave the pieces in [Part 3](#), while (3) shows the names and ids of all the pieces you made in [Part 4](#). Next, choose the animation in (2) that you will begin working on. Know that you do not have to create an animation for burrowing unless you intend to make your unit a zombie unit. Once you have selected an animation, your unit will appear in (4), known as the viewer window.



To start animating, click the add line button next to (6) and a new line will appear.

part id	modification	loop	name
1	5	-1	

Part ID - What part you are trying to animate
Enter the part id of the part you are trying to animate. If I wanted to animate CatMoj's head, then I would choose enter 2, which I can reference from (3).

Modification - What you are doing with that part
(5) shows a list of all of the modification you can do to a part. These are exactly the same as the ones in [Part 4](#). (Except for extend which I have no idea what it does.)

Loop - Whether a part of an animation loops or not
-1 is an infinite loop that won't stop unless the unit switches to a different animation. 1 will not loop and will wait for all of the other animations to finish before it restarts.

Name - It's a name
What did you expect, it's a name

Each line manipulates a part a certain way, but you can have multiple lines affecting one part.

part id	modification	loop	name
2	8	-1	Scale - Head
2	11	-1	Rotate - Head

For example, these 2 lines allow the head to be scaled and rotated at the same time.

Let's start writing out animation lines. I will use this line I made in (6) as an example. I will be scaling CatMoji's Head

part id	modification	loop	name
2	8	-1	Scale - Head

Go to (8) and click add line to create a new line. You might see a part disappear or do other weird things, but this is fine.

frame	value	easing	parameter
0	0	0	0
10	1000	0	0

Frame - Represents what the state of a part is at a certain frame. (Ex: At frame 0, the part is at a scale of 0, or non-existent. At frame 10, the part is at a scale of 1000, or normal scale.)

Value - How much a part is changing

This is different for each Modification. (Ex: For Scale, value changes size, but for Sprite, value is what Sprite ID, from Part 3, a piece will change to.)

Easing - Rate of animation

0 - Linear Ease - The animation will flow at a constant pace (Ex: If an object is moving, it will move at the same speed the entire time)

1 - No easing - Basically skips that line

~~Don't use it. It's stupid~~

** it does have a purpose for specific effects**

3 - Sine Easing - Speed up at beginning at end

Sine easing slows down at the beginning, speeds up to normal speed in the middle, and slows down at the end.

This is useful for making some animations look more “lifelike” or fluid.

2 - Ease in or Ease out - Slower at beginning or or end
The animation will slow down at the beginning and then speed up to a normal speed or start at a normal speed and slow down at the end. This is the only type of easing that uses “Parameter”.

Parameter - Rate of Ease In/Out

Negative Numbers - Ease In

Positive Numbers - Ease Out

Having a number closer to 0 makes the easing happen sooner and slower

Remember to put your easing modifications on the line before the frame you want to ease to.

frame	value	easing	parameter
0	1000	2	-5
50	10000	0	0



Let's animate Catmoji Growing and Changing Sprites

part id	modification	loop	name
2	8	-1	Scale - Head

frame	value	easing	parameter
0	1000	3	0
50	3000	0	0

So, I start with a new modification line set to Part ID 2, and a modification of 8 to scale the head. I ignore changing the loop because I don't need this to stop during the animation.

I then create a new frame line set at frame zero and a value of 1000 to set the size of Catmoji's head at neutral during the beginning of the animation. Setting easing to 3 lets the animation have an ease in and out

I then create a new line for the next part of the animation. I set frame to 50 and value to 3000 so that between frames 0 and 50 CatMoji's head scales from a value of 1000 to 3000.

Now, I make a new modification line with the modification of 2, so we can change the head from a normal head to an attack head.

part id	modification	loop	name
2	8	-1	Scale - Head
2	2	-1	Sprite - Head

frame	value	easing	parameter
0	1	0	0
60	2	0	0
90	2	0	0

I like to add a line at frame zero with no change in value for Modification Lines that don't begin at frame 0.

At frame 60, I set the value to 2 so that the sprite can change from 1 which is labeled as "Head Neutral" to 2 which is labeled as "Attack Head 1". This results in the CatMoji's head opening after it finishes growing. I then add another Frame Line at frame 90 to have Catmoji hold that face for 30 frames since this is the last Frame Line in the animation.

part id	modification	loop	name
2	8	-1	Scale - Head
2	2	-1	Sprite - Head

frame	value	easing	parameter
0	1000	3	0
50	3000	0	0
90	3000	0	0

Since we did that, I need to go back to the Scale Modification Line and set a new Frame Line at Frame 90 so that the animation does not desync.

Now you have learned how to animate.



Have fun